DEP 2: Data Exploration

FIT5147

Data Exploration and Visualization

SALDI

33780331

Table of Content

Introduction

Data Wrangling

Data Checking

Data Exploration and Visualization

Conclusion

Reflection

Reference

Appendix

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Introduction

Problem Description

Recently, more media outlets have highlighted how Micro, Small, and Medium Enterprises (MSMEs) in Indonesia form an important component of the country's economic growth [1,2]. In my opinion, this significantly impacts the Gross Domestic Product (GDP) and inflation. Furthermore, I am interested in learning how well Indonesian MSMEs are growing, as the repayment loans indicate.

Question

1. Based on each MSMEs metric, how does the trend depict the productivity level of MSMEs in Indonesia?
2. Which industrial origin has the highest credit realization, and how does it impact the GDP deflator and inflation rate?

Motivation

My prior experience as a banker reinforced my desire to become a financial analyst. While working, I encountered numerous creditors whose repayments remained unpaid. I have learned that this will affect the GDP's growth and the inflation rate.

Data Wrangling

Data Sources

The data used is sourced from the Central Bank of Indonesia;

1. **Indicators of Micro, Small, and Medium Enterprises (MSMEs) 2016-2024**, reported by the Bank Indonesia through Statistics of Indonesia's Financial System (URL: https://www.bi.go.id/id/statistik/ekonomi-keuangan/sski/Pages/SSKI_Juli_2024.aspx). The data variables contain outstanding MSMEs credit, number of account credit banks, growth of MSMEs credit, and NPL MSMEs credit.
2. Outstanding Micro, Small, and Medium Enterprises Credits of Commercial Banks (MSMEs), Gross Domestic Product Deflator, and Inflation of Selected Economies 2020-2024. The Bank of Indonesia has reported it in the Indonesian Economic and Financial Statistics (URL: https://www.bi.go.id/en/statistik/ekonomi-keuangan/seki/Default.aspx#headingOne). The data variables have outstanding MSMEs credit of commercial banks, categorized by industrial origin, inflation rates, and gross domestic product (GDP) deflator. All the data are ordered by year from 2016 to 2024.

Data Cleaning

Some data that are not necessarily included have been removed from the dataset using Microsoft Excel. There are four datasets; then, the data are joined into one file with three sheets that use year as connections. This project only focuses on commercial banks, so the other instruments irrelevant to the field, such as rural banks (conventional and sharia), are filtered.

Data Transformation

The crucial step is modifying the data structure to make it readable for programming activities in the analysing process. The years are transformed into a list, and the same actions are also

implemented in enterprise classifications. After ordering those variables, the other instruments, which contain the value of each classification, are transferred and then transposed from rows to columns using a special copy-paste feature called transpose.

There are two datasets with the same purpose to show the particular value based on sector loan recipients.

By Main Sector Loan Recipients
Wholesale and Retail Sector
Manufacturing Sector
Agriculture, Hunting and Forestry Sector
Transportation, Storage and Communication Sector
Financial Intermediary Sector
Real Estate, Leasing and Business Services Sector
Construction, Mining, and Quarrying Sector
Electricity, Gas and Water Sector
Accomodation, Food and Beverage Services

Figure 1: Table Indicator of MSMEs

By Industrial Origin
Agriculture, Forestry and Fishery
Mining and Quarrying
Manufacturing
Electricity and Gas Supply
Water Supply, Sewerage, Waste Management and Remediation Activities
Construction
Wholesale and Retail Trades, Repair of Motor Vehicles and Motorcycles
Transport and Storage
Accomodation and Food Service Activities
Information and Communication
Financial and Insurance Services
Real Estate Activities
Business Services
Public Administration and Defence; Compulsory Social Security
Education
Human Health and Social Work Activities
Other Services Activities

Figure 2: Table Gross Domestic Product Deflator

Those variables should be uniform in the analysis process to avoid conflict for each category. Therefore, the data from Figure two are converted the same as Figure 1 by calculating the value, which has the same color, using a sum function in Excel.

Data Checking

Year	Main Factor Loan Recipients	Total of Outstanding Amount (Billions of Rp)	Growth of Credit (%, yoy)	GDP Deflator (%)
2016	Others Sector	60,950		11.13
2017	Others Sector	76,834		12.68
2018	Others Sector	78,621		39.96
2019	Others Sector	85,612		32.18
2020	Others Sector	87,663		49.17
2021	Others Sector	100,719		82.14
2022	Others Sector	111,150		68.48
2023	Others Sector	125,972		110.73
2024	Others Sector	128,709		143.52

Figure 3: Missing Values

After restructuring the data format, there are missing values in the credit growth where other sectors do not have value. Rather than cutting the data, we could find the value by calculating the formula $Growth\ Rate\ = \frac{(Previous\ Year\ Outstanding - Current\ Year\ Outstanding)}{Current\ Year\ Outstanding} \times 100$ in Excel.

Year	Main Factor Loan Recipients	Total of Outstanding Amount (Billions of Rp)	Growth of Credit (%, yoy)	GDP Deflator (%)
2016	Others Sector	60,950		11.13
2017	Others Sector	76,834	26.06	12.68
2018	Others Sector	78,621	2.33	39.96
2019	Others Sector	85,612	8.89	32.18
2020	Others Sector	87,663	2.39	49.17
2021	Others Sector	100,719	14.89	82.14
2022	Others Sector	111,150	10.36	68.48
2023	Others Sector	125,972	13.34	110.73
2024	Others Sector	128,709	2.17	143.52

Figure 4: Missing Values

However, we cannot find the first value of other sectors because there is no information from the previous year in 2015. To deal with this issue, a box plot can be used to assume the value.

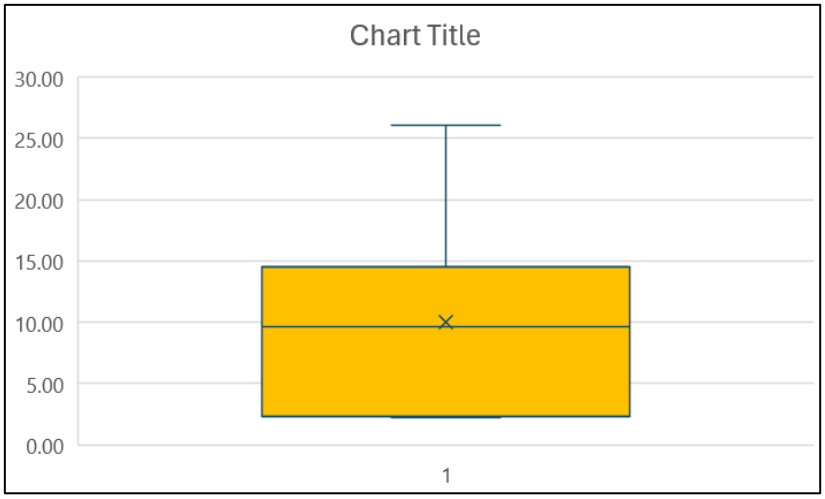

Figure 5: Growth Credit Boxplot

From the boxplot, we added the value at 20.50 with the assumption following the same patterns of each value where every three years has adjacent patterns. That value also meets the requirements as not indicated as an outlier.

Data Exploration and Visualization

Enterprises Productivity Trend

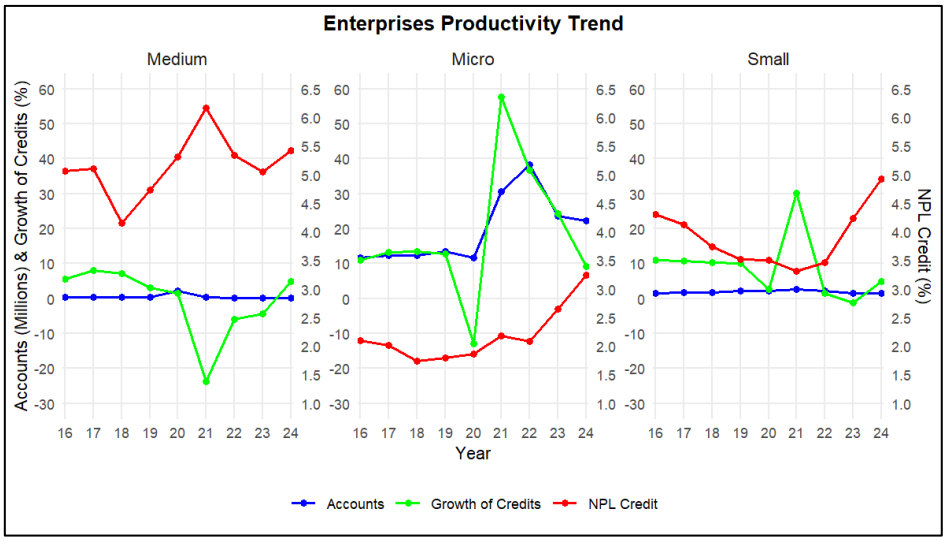

Figure 6: Enterprises Productivity Trend

The graph presents a comprehensive view of the productivity trends for three categories of enterprises – Medium, Micro, and Small – from 2016 to 2024. The analysis focuses on three key indicators: the number of accounts (in millions), the growth of credits (percentage), and non-performing loans (NPL) credit (percentage). These indicators provide insights into these enterprises' financial health and credit behavior over time.

Medium Enterprises: A Slow and Steady Path

Medium enterprises exhibit relatively stable behavior over the observed period. **The number of accounts** remains consistent mainly, indicating that the scale of these businesses has not experienced significant fluctuations. However, **the growth of credits** demonstrates moderate variability, with a decline in the middle of the timeline, followed by a slight recovery. This suggests that credit growth has been conservative, reflecting medium enterprises' cautious approach to expanding their financial commitments.

Nevertheless, **the NPL credit** reveals a gradual but concerning increase. Beginning in 2019, the percentage of non-performing loans started to climb, peaking in 2021. This trend signals growing financial pressure as these enterprises encounter difficulties managing debt obligations, potentially due to external economic factors or internal inefficiencies.

Micro Enterprises: A High-Risk, High-Reward Scenario

Micro Enterprises, in contrast, show more volatile trends. **The number of accounts** remains relatively stable, slightly increasing in later years. However, **the growth of credits** experienced a dramatic spike in 2021, reaching its highest point during this period. This indicates a period of aggressive expansion, where micro-enterprises took on significant credit to fuel growth. Yet, this boom is short-lived, as the growth rate plummets soon after.

This boom-and-bust cycle is reflected in **the NPL credit**, which rises sharply following the peak in credit growth. By 2024, micro-enterprises face a substantial increase in non-performing loans, suggesting that the rapid credit expansion may have led to unsustainable debt levels. The high volatility in both credit growth and NPLs indicates a high-risk, high-reward dynamic within this category of businesses.

Small Enterprises: Stability Amid Fluctuations

Small enterprises display a more balanced trajectory over time. **The number of accounts** remains relatively stable, with slight variation throughout the period. However, **the growth of credits** fluctuates, particularly in 2022, where there is a notable decline followed by a recovery. Despite these fluctuations, credit growth in small enterprises remains within a moderate range.

The NPL credit shows a clear downward trend between 2016 and 2020, suggesting an initial improvement in the financial health of small enterprises, possibly due to better credit management or favorable economic conditions. However, after 2020, NPLs start rising again, sharply increasing by 2023 and continuing through 2024. By 2024, the NPL percentage exceeded 5%, indicating that many small enterprises struggled to meet their loan obligations as economic conditions worsened.

Distribution Patterns and Financial Insights from Industry Loan Data

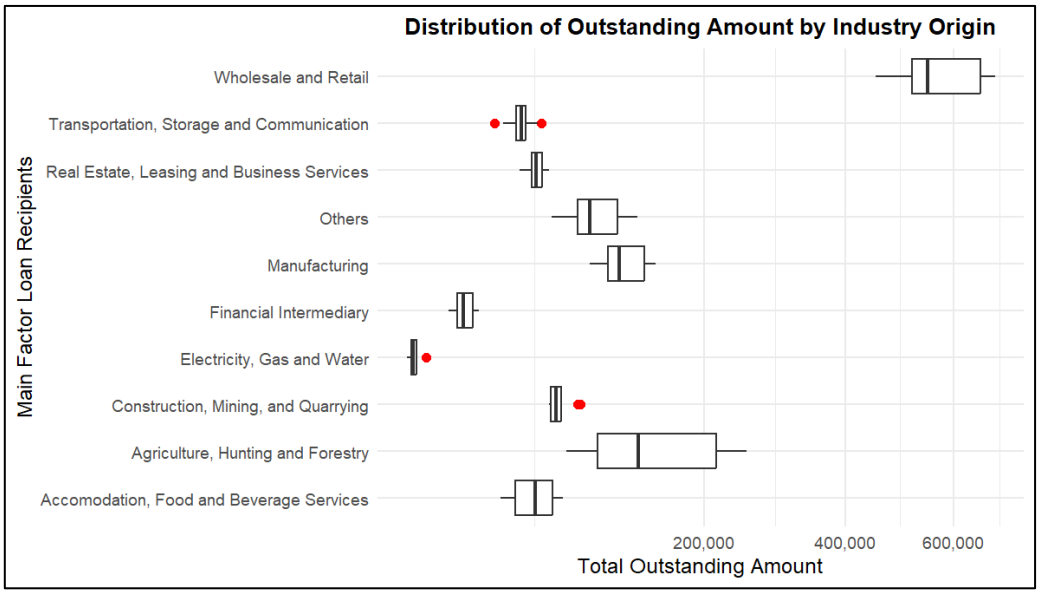

Figure 7: Distribution of Outstanding Among by Industrial Origin

The boxplot visualization illustrates the distribution across different industries, providing insights into variability, central tendency, and the presence of outliers within each sector. Each boxplot represents a distinct industry, with the total outstanding loan amount on the x-axis and industry categories on the y-axis. The width of the boxes, along with the whiskers, represents the interquartile range (IQR), which clearly indicates the spread and skewness of the loan amount.

The visualization highlights the varying levels of loan distributions across different industries. Sectors such as **Wholesale and Retail** and **Real Estate, Leasing, and Business Services** display large loan amounts and wider variability, reflecting their scale and potentially greater financial needs. In contrast, sectors like **the Financial Intermediary** and **Electricity, Gas, and Water** show tighter distributions, suggesting lower loan sizes. The presence of outliers in several industries, particularly **Transportation, Storage, and Communication, Construction, Mining, and Quarrying,** and **Electricity, Gas and Water,** indicates the existence of businesses with loan amounts significantly deviating from industry norms, which could reflect different business models or financial conditions within those industries.

Year	Main Factor Loan Recipients	Total of Outstanding Amount (Billions of Rp)
2016	Transportation, Storage and Communication	29461.000
2018	Construction, Mining, and Quarrying	78986.357
2019	Construction, Mining, and Quarrying	80583.035
2019	Electricity, Gas and Water	6669.012
2024	Transportation, Storage and Communication	53934.499

Figure 8: The Outlier of Total Outstanding by Sectors

These outliers highlight periods of intense financial activity or large investments within specific sectors, likely driven by infrastructure development, industry expansion, or government initiatives. Each outlier represents a deviation from the norm, indicating significant fluctuations in financial support for these industries during the respective years.

Annual Credit Distribution Insights

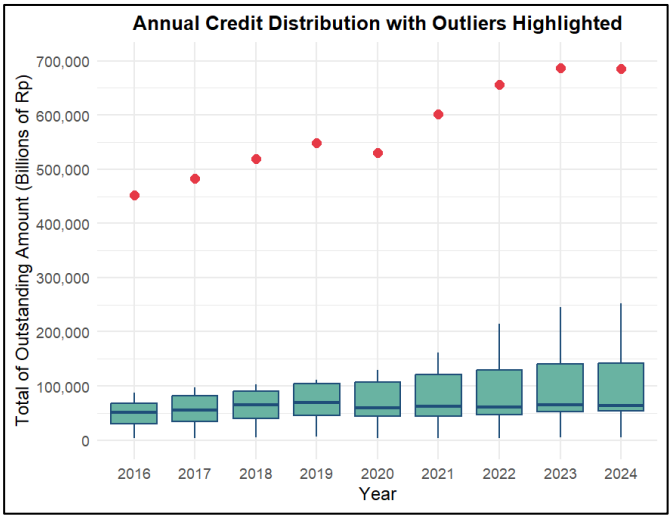

Figure 9: Annual Credit Distribution with Outliers Highlighted

The boxplot illustrates a steady increase in total outstanding credit from 2016 to 2024, with greater variability and a rise in median values after 2020. Significant outliers, marked by red points, indicate that specific industries experienced disproportionately higher credit amounts, particularly in recent years. This suggests overall growth in financial activity and notable sectoral differences in credit distribution. The expanding range in later years highlights increasing disparities in credit needs across industries.

Year	Main Factor Loan Recipients	Total of Outstanding Amount (Billions of Rp)
2016	Wholesale and Retail	451755.0
2017	Wholesale and Retail	482635.5
2018	Wholesale and Retail	518626.7
2019	Wholesale and Retail	548284.8
2020	Wholesale and Retail	530668.6
2021	Wholesale and Retail	601408.0
2022	Wholesale and Retail	656431.3
2023	Wholesale and Retail	686505.3
2024	Wholesale and Retail	684784.2

Figure 10: The Outlier by Sector

The wholesale and retail sector experienced significant and consistent growth in outstanding credit amounts from 2016 to 2024. Its role as a critical driver of the economy, along with possibly increasing reliance on credit to finance expansion and innovation, has

made it a consistent outlier in terms of financial activity. The large amounts reflect the sector's growing demands and pivotal role in sustaining economic growth.

Interrelationship Between Credit Growth, Total Outstanding Credit, and Inflationary Pressures

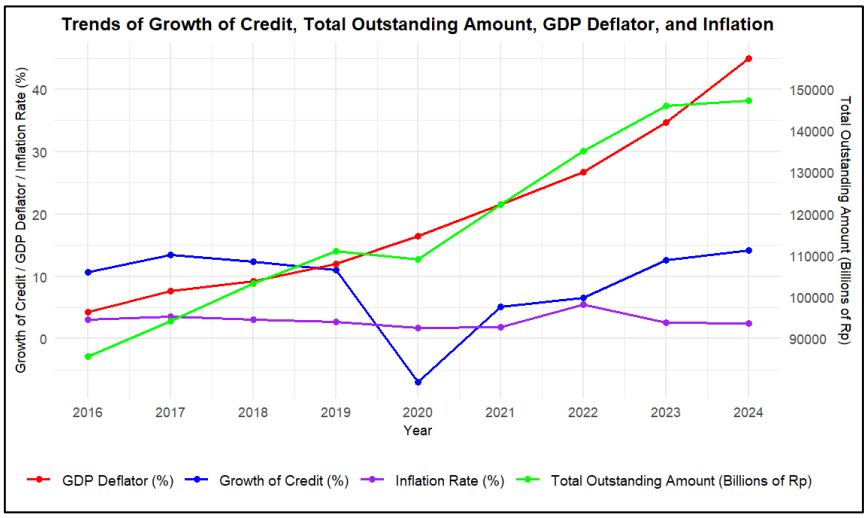

Figure 11: Relationship of Financial Instruments

The graph presents a complex interplay between credit growth, inflation, and overall economic pricing dynamics. The decline in **the growth of credit** in 2020 likely reflects economic contractions and financial market volatility during the COVID-19 pandemic. Conversely, the steady increase in **the total outstanding amount** suggests resilience in borrowing trends, possibly driven by accommodative fiscal and monetary policies.

The GDP Deflator's upward trajectory highlights rising prices in the broader economy, possibly reflecting supply-side constraints, increased production costs, or higher demand. Interestingly, **the inflation rate**, which captures consumer price changes, remains stable, signaling potential differences in price dynamics between domestic production and consumer prices.

This visualization underscores the importance of monitoring multiple financial and economic indicators to capture a holistic view of macroeconomic conditions, especially during economic uncertainty and recovery periods.

The Relationship Between Growth of Credit and GDP Deflator

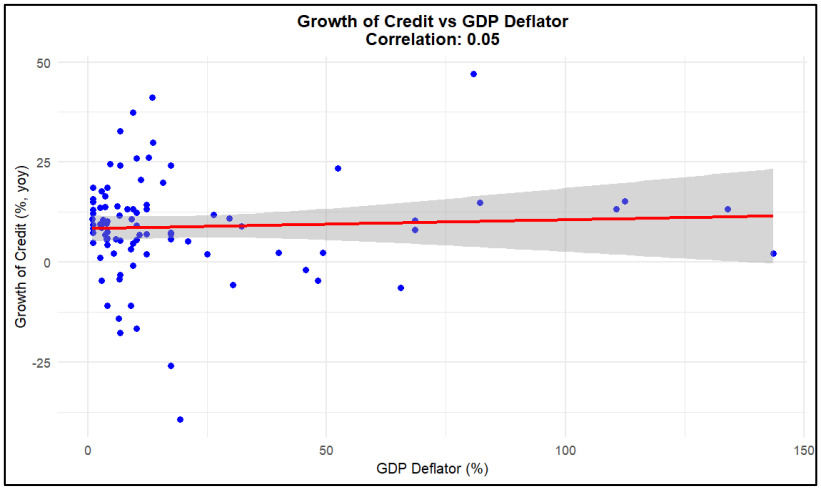

Figure 12: Correlation Between Growth of Credit and GDP Deflator

The scatterplot illustrates the relationship between the **Growth of Credit** and the **GDP Deflator**, with a linear regression line (red) and a confidence interval (gray). Most of the data points are concentrated around lower GDP deflator values (below 50%), with credit growth ranging from -45% to 50%, mostly clustered between 0% and 25%. The nearly flat regression line suggests no strong trend between the two variables, as confirmed by a weak correlation coefficient of 0.05, indicating little to no linear association. The widening confidence interval at higher GDP deflator values highlights increased uncertainty, likely due to fewer observations in that range. Overall. The analysis indicates that the GDP deflator does not significantly affect credit growth.

The Relationship Between Growth of Credit and Inflation

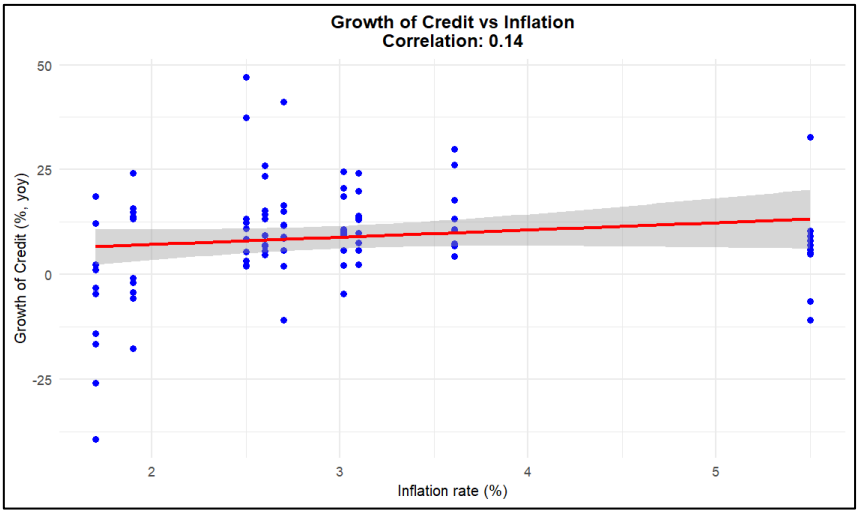

Figure 13: Correlation Between Growth of Credit and Inflation

The scatterplot shows the relationship between the **Growth of Credit** and the **Inflation Rate**, with blue points representing data observations. Inflation rates range from 1.5% to 5.5%, while credit varies from -40% to 50%. The red regression line has a slight upward slope, indicating a weak positive association between inflation and credit growth—the correlation coefficient of 0.14 confirms this weak relationship, suggesting that while credit growth may slightly increase with inflation, the correlation is not strong enough to be meaningful. The grey confidence interval widens at higher inflation rates, reflecting increased variability, further suggesting that inflation is not a reliable predictor of credit growth. Overall, the weak positive correlation implies that inflation has a minimal impact on credit growth.

Conclusion

Based on data exploration, here are the conclusions that address the two original research questions:

1. **Productivity Trends of MSMEs in Indonesia:** The trends indicate varied performance across different sizes of enterprises. Medium Enterprises showed stability in account numbers but faced a rise in non-performing loans (NPL) over time. Micro-enterprises experienced volatile credit growth, peaking in 2021, but this led to a sharp increase in NPLs in the next period, indicating unsustainable debt levels. Small enterprises displayed moderate fluctuations in credit growth and a worrying rise in NPLs after 2020. The productivity trend suggests that while MSMEs have grown, their financial health, especially regarding debt repayment, has deteriorated over time.
2. **Credit Realization by Industrial Origin and Its Impact on GDP and Inflation:** Industries such as Wholesale and Retail, and Real Estate showed higher credit

realization, reflecting their larger financial needs. However, the analysis of the relationship between credit growth and GDP deflator revealed no significant correlation, suggesting that credit growth does not directly impact inflation. Similarly, there was only a weak positive relationship between credit growth and inflation, indicating that while there might be some inflationary pressure due to credit growth, the effect is not substantial.

In summary, while credit expansion has supported MSMEs' growth, the financial risk associated with increasing NPLs highlights potential issues, especially for smaller enterprises. Additionally, the weak link between financial instruments suggests that credit policies may have limited direct influence on macroeconomic inflationary trends.

Reflections

In reflecting on this project, I learned several key lessons:

1. **Data Integrity and Preparation:** One of the biggest challenges was ensuring data accuracy and dealing with missing values. This experience reinforced the importance of data cleaning and transformation in generating reliable insights. In hindsight, I would have further allocated more time to investigate the missing values and considered alternative methods, such as more advanced imputation, to handle the gaps more effectively.
2. **Understanding of MSMEs Financial Trends:** Analysing the trends in MSMEs productivity and credit growth deepened my understanding of the financial dynamics in these sectors. The volatility observed, particularly in the micro-enterprise sector, highlighted the need for risk management strategies in credit distribution.
3. **Economic Indicators Interpretation:** While I expected a stronger relationship between financial instruments, the weak correlation found in this project taught me that macroeconomic trends are influenced by a multitude of factors beyond just credit growth. In the future, I would broaden the analysis by incorporating other economic indicators, such as unemployment rates or government fiscal policies, to gain a more holistic understanding.
4. **Improving Visualization:** The visualization process was insightful, but looking back, I might have experimented with different visualization techniques or tools to represent better certain complex relationships, such as those between credit growth and economic variables. This could have made the insights more accessible to non-technical stakeholders.

Overall, this project provided valuable insights, but more refined approaches to data handling and broader analytical perspectives could improve the results in future analyses.

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APPENDIX

R code creates Figure 6

```r
library(ggplot2)
library(readxl)
library(dplyr)

# Reading data from the Excel file
df_enterprises <- read_excel("DEP2a.xlsx", sheet = "Enterprises Classifications")

# Remove leading spaces from column names (if necessary)
colnames(df_enterprises) <- trimws(colnames(df_enterprises))

# Convert Year column to numeric format for easier handling
df_enterprises$Year <- as.numeric(df_enterprises$Year)

# Manually set limits for primary and secondary axes
primary_min <- -30
primary_max <- 60
npl_min <- 1
npl_max <- 6.5

# Calculate the scaling factor for secondary axis based on primary axis range
scale_factor <- (primary_max - primary_min) / (npl_max - npl_min)

# Plotting with facet_wrap for all classifications
p_all <- ggplot(df_enterprises, aes(x = Year)) +
  # Lines with Points (Dots) for Accounts
  geom_line(aes(y = `Number of Account (in Millions)`, color = "Accounts"), size = 0.8) +
  geom_point(aes(y = `Number of Account (in Millions)`, color = "Accounts"), size = 1.5) +

  # Lines with Points (Dots) for Growth of Credits
  geom_line(aes(y = `Growth of Credits (%, yoy)`, color = "Growth of Credits"), size = 0.8) +
  geom_point(aes(y = `Growth of Credits (%, yoy)`, color = "Growth of Credits"), size = 1.5) +

  # Lines with Points (Dots) for NPL Credit
  geom_line(aes(y = (`NPL Credit (%)` - npl_min) * scale_factor + primary_min, color = "NPL Credit"), size = 0.8) +
  geom_point(aes(y = (`NPL Credit (%)` - npl_min) * scale_factor + primary_min, color = "NPL Credit"), size = 1.5) +

  # Primary axis for Accounts and Growth of Credits
  scale_y_continuous(
    name = "Accounts (Millions) & Growth of Credits (%)",
    limits = c(primary_min, primary_max),
    breaks = seq(primary_min, primary_max, by = 10),

    # Secondary axis for NPL Credit (%)
    sec.axis = sec_axis(~ (. - primary_min) / scale_factor + npl_min, name = "NPL Credit (%)", breaks = seq(npl_min, npl_max, by = 0.5))
  ) +

  # Set uniformly spaced year intervals on the x-axis
  scale_x_continuous(breaks = unique(df_enterprises$Year),labels = function(x) substring(as.character(x), 3, 4), expand = c(0.01, 0.01)) +

  # Facet wrap to create separate plots for each classification
  facet_wrap(~ `Enterprises Classifications`, nrow = 1, scales = "free_y") +

  labs(title = "Enterprises Productivity Trend", x = "Year") +
  scale_color_manual(values = c("Accounts" = "blue", "Growth of Credits" = "green", "NPL Credit" = "red")) +

  # Adjust theme to match the desired layout
  theme_minimal(base_size = 10) +  # Decrease the base font size
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 12),
    axis.title.y = element_text(size = 10),
    axis.title.y.right = element_text(size = 10),
    axis.title.x = element_text(size = 10),
    legend.position = "bottom",
    legend.title = element_blank(),
    legend.text = element_text(size = 8),
    strip.text = element_text(size = 10),
    axis.text = element_text(size = 8),
    panel.grid.minor = element_blank()
  )

# Display the plot
p_all
```

R code creates Figure 7 and Figure 8

```r
# Load required libraries
library(tidyverse)
library(ggplot2)
library(readxl)

# Reading the data from Excel file (replace 'your_file_path.xlsx' with the actual file path)
file_path <- "DEP2a.xlsx"

# Load the data into a data frame
df_industry_origin <- read_excel(file_path, sheet = "Industry Origin")

# Create a box plot with sqrt transformation on the y-axis and labels rotated 90 degrees
ggplot(df_industry_origin, aes(x = `Main Factor Loan Recepients`, y = `Total of Outstanding Amount (Billions of Rp)`)) +
  geom_boxplot(outlier.colour = "red", outlier.shape = 16, outlier.size = 2, notch = FALSE) +
  labs(title = "Distribution of Outstanding Amount by Industry Origin",
       x = "Main Factor Loan Recipients",
       y = "Total Outstanding Amount") +
  scale_y_sqrt(labels = scales::comma) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 12, face = "bold")
  ) +
  coord_flip()

# Calculate quartiles and IQR to identify outliers
outlier_data <- df_industry_origin %>%
  group_by(`Main Factor Loan Recepients`) %>%
  summarise(
    Q1 = quantile(`Total of Outstanding Amount (Billions of Rp)`, 0.25),
    Q3 = quantile(`Total of Outstanding Amount (Billions of Rp)`, 0.75),
    IQR = Q3 - Q1,
    lower_bound = Q1 - 1.5 * IQR,
    upper_bound = Q3 + 1.5 * IQR
  )

# Identify outliers for each category including the year
outliers <- df_industry_origin %>%
  inner_join(outlier_data, by = "Main Factor Loan Recepients") %>%
  filter(`Total of Outstanding Amount (Billions of Rp)` < lower_bound |
           `Total of Outstanding Amount (Billions of Rp)` > upper_bound) %>%
  select(Year, `Main Factor Loan Recepients`, `Total of Outstanding Amount (Billions of Rp)`)

# Print the outliers with their specific year
View(outliers)
```

R code creates Figure 9 and Figure 10

```r
# Load necessary libraries
library(ggplot2)
library(dplyr)
library(readxl)

# Load the data from the second sheet
data <- read_excel("DEP2a.xlsx", sheet = 2)

# Calculate IQR and identify outliers
outliers <- data %>%
  group_by(Year) %>%
  mutate(IQR = IQR(`Total of Outstanding Amount (Billions of Rp)`),
         lower_bound = quantile(`Total of Outstanding Amount (Billions of Rp)`, 0.25) - 1.5 * IQR,
         upper_bound = quantile(`Total of Outstanding Amount (Billions of Rp)`, 0.75) + 1.5 * IQR,
         is_outlier = `Total of Outstanding Amount (Billions of Rp)` < lower_bound | `Total of Outstanding Amount (Billions of Rp)` > upper_bound) %>%
  filter(is_outlier)

# Print outliers
View(outliers)

# Create the boxplot with improved color combinations
ggplot(data, aes(x = as.factor(Year), y = `Total of Outstanding Amount (Billions of Rp)`)) +
  geom_boxplot(fill = "#69b3a2", color = "#1f4e79") +  # Use teal for the boxes and dark blue for borders
  geom_point(data = outliers, aes(x = as.factor(Year), y = `Total of Outstanding Amount (Billions of Rp)`), color = "#e63946", size = 2.5) +
  labs(title = "Annual Credit Distribution with Outliers Highlighted",
       x = "Year",
       y = "Total of Outstanding Amount (Billions of Rp)") +
  scale_y_continuous(limits = c(0, 700000), breaks = seq(0, 700000, by = 100000), labels = scales::comma) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 12, face = "bold"
    ))
```

R code creates Figure 11

```r
# Load necessary libraries
library(tidyverse)
library(readxl)
library(ggplot2)

# Load the data directly from the Excel file
trend_data <- read_excel("DEP2a.xlsx", sheet = "Industry Origin")
inflation_data <- read_excel("DEP2a.xlsx", sheet = "Inflation")

# Merging inflation data with the main data
trend_data <- trend_data %>%
  left_join(inflation_data, by = "Year")

# Grouping the data by 'Year' and calculating mean values for each metric
trend_data_summary <- trend_data %>%
  group_by(Year) %>%
  summarise(
    Growth_of_Credit = mean(`Growth of Credit (%, yoy)`, na.rm = TRUE),
    Total_Outstanding_Amount = mean(`Total of Outstanding Amount (Billions of Rp)`, na.rm = TRUE),
    GDP_Deflator = mean(`GDP Deflator (%)`, na.rm = TRUE),
    Inflation_Rate = mean(`Inflation rate`, na.rm = TRUE)
  )

# Manually adjust the scaling factor so that 90,000 is aligned with 0 and 150,000 is aligned with 40
scaling_factor <- (150000 - 90000) / 40  # Align the secondary axis range with primary axis range

# Create the dual-axis plot
p <- ggplot(trend_data_summary, aes(x = Year)) +
  geom_line(aes(y = Growth_of_Credit, color = "Growth of Credit (%)"), size = 0.8) +
  geom_point(aes(y = Growth_of_Credit, color = "Growth of Credit (%)"), size = 1.5) +
  geom_line(aes(y = GDP_Deflator, color = "GDP Deflator (%)"), size = 0.8) +
  geom_point(aes(y = GDP_Deflator, color = "GDP Deflator (%)"), size = 1.5) +
  geom_line(aes(y = Inflation_Rate, color = "Inflation Rate (%)"), size = 0.8) +
  geom_point(aes(y = Inflation_Rate, color = "Inflation Rate (%)"), size = 1.5) +
  geom_line(aes(y = (Total_Outstanding_Amount - 90000) / scaling_factor, color = "Total Outstanding Amount (Billions of Rp)"), size = 0.8) +
  geom_point(aes(y = (Total_Outstanding_Amount - 90000) / scaling_factor, color = "Total Outstanding Amount (Billions of Rp)"), size = 1.5) +
  scale_y_continuous(
    name = "Growth of Credit / GDP Deflator / Inflation Rate (%)",
    sec.axis = sec_axis(~ . * scaling_factor + 90000, name = "Total Outstanding Amount (Billions of Rp)",
                        breaks = seq(90000, 150000, by = 10000))
  ) +
  scale_x_continuous(breaks = seq(min(trend_data_summary$Year), max(trend_data_summary$Year), by = 1)) +
  scale_color_manual(values = c("Growth of Credit (%)" = "blue",
                                "GDP Deflator (%)" = "red",
                                "Inflation Rate (%)" = "purple",
                                "Total Outstanding Amount (Billions of Rp)" = "green")) +
  labs(title = "Trends of Growth of Credit, Total Outstanding Amount, GDP Deflator, and Inflation", color = NULL) +
  theme_minimal() +
  theme(
    legend.position = "bottom",  # Adjust legend position like in the image
    plot.title = element_text(hjust = 0.5, size = 11, face = "bold"),  # Centered title
    axis.title = element_text(size = 8),  # Adjust the font size of axis titles
    axis.text = element_text(size = 8)     # Adjust the font size of axis tick labels
  )

# Print the plot
print(p)
```

R code creates Figure 12 and Figure 13

```r
library(readxl)
library(ggplot2)

# Read the Excel file
file_path <- "DEP2a.xlsx"
df_sheet2 <- read_excel(file_path, sheet = 2)
df_sheet3 <- read_excel(file_path, sheet = 3)

# Calculate the correlation between Growth of Credit and GDP Deflator
correlation <- cor(df_sheet2$`Growth of Credit (%, yoy)`, df_sheet2$`GDP Deflator (%)`)
print(paste("Correlation:", correlation))

# Merge the two datasets by 'Year'
merged_data <- merge(df_sheet2, df_sheet3, by = "Year")

# Calculate the correlation between Growth of Credit and Inflation
correlation_inflation <- cor(merged_data$`Growth of Credit (%, yoy)`, merged_data$`Inflation rate`)
print(paste("Correlation between Growth of Credit and Inflation:", correlation_inflation))

# Plot the scatterplot with a linear regression line
ggplot(df_sheet2, aes(x = `GDP Deflator (%)`, y = `Growth of Credit (%, yoy)`)) +
  geom_point(color = "blue") +
  geom_smooth(method = "lm", color = "red") +
  labs(title = paste("Growth of Credit vs GDP Deflator\nCorrelation:", round(correlation, 2)),
       x = "GDP Deflator (%)",
       y = "Growth of Credit (%, yoy)") +
  theme_minimal()+
  theme(
    plot.title = element_text(hjust = 0.5, size = 11, face = "bold"),
    axis.title.x = element_text(size = 9),
    axis.title.y = element_text(size = 9),
    axis.text = element_text(size = 8))

# Plot the scatterplot with a linear regression line
ggplot(merged_data, aes(x = `Inflation rate`, y = `Growth of Credit (%, yoy)`)) +
  geom_point(color = "blue") +
  geom_smooth(method = "lm", color = "red") +
  labs(title = paste("Growth of Credit vs Inflation\nCorrelation:", round(correlation_inflation, 2)),
       x = "Inflation rate (%)",
       y = "Growth of Credit (%, yoy)") +
  theme_minimal()+
  theme(
    plot.title = element_text(hjust = 0.5, size = 11, face = "bold"),
    axis.title.x = element_text(size = 9),
    axis.title.y = element_text(size = 9),
    axis.text = element_text(size = 8))
```